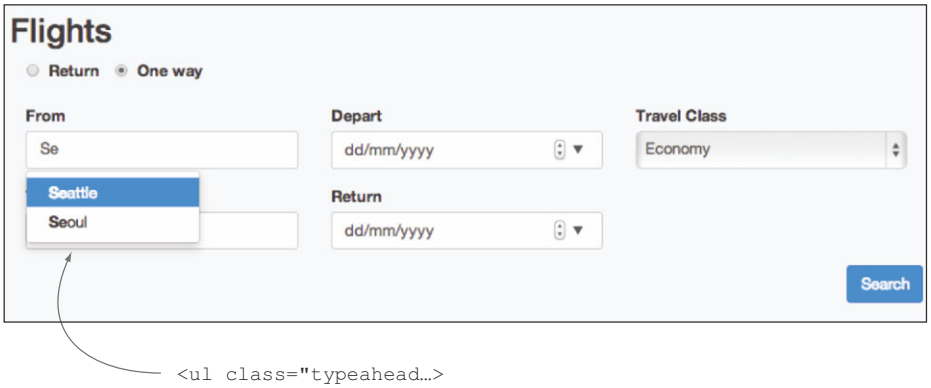


Figure 8.11 Type-ahead fields are a common example of asynchronous lookups.

directly, without reloading. When this happens, WebDriver will not know if or when it needs to wait for updates, which may cause the test to fail unexpectedly.

Fortunately, it's relatively easy to tell WebDriver when it needs to wait and what it should wait for. Your main tools for achieving this are the `Wait` interface and its implementations.

Figure 8.11 illustrates a common case where waiting may be useful. On the Frequent Flyer booking page, the From and To fields are configured with a type-ahead capability that displays matching cities in a drop-down list as the user types.

The time it takes to display the drop-down list can be unpredictable, as it may depend on the speed of the network and the target server. Depending on the nature of the web page, there are several ways you can handle this sort of delay.

USING THE IMPLICIT WAIT

By default, if WebDriver doesn't find a web element, it will fail immediately. But this behavior is configurable. For example, if you wanted WebDriver to wait for five seconds before declaring forfeit, you could set the implicit wait time to five seconds, as shown here:

```
driver.manage().timeouts().implicitlyWait(5, TimeUnit.SECONDS);
```

This is a blunt tool, as it will apply every time WebDriver looks up an element, which can slow down the tests in other places. A more refined approach is to use *explicit waits*.

USING EXPLICIT WAITS

WebDriver also lets you wait for specific events. The easiest way to do this is to use the `ExpectedConditions` class. This class provides a large number of useful predefined conditions that make the API more convenient to use. These include waiting for elements to be present (or not present), visible (or invisible), clickable, and so forth. For example, the following code waits until the type-ahead list (identified here by its class name) is present on the screen: